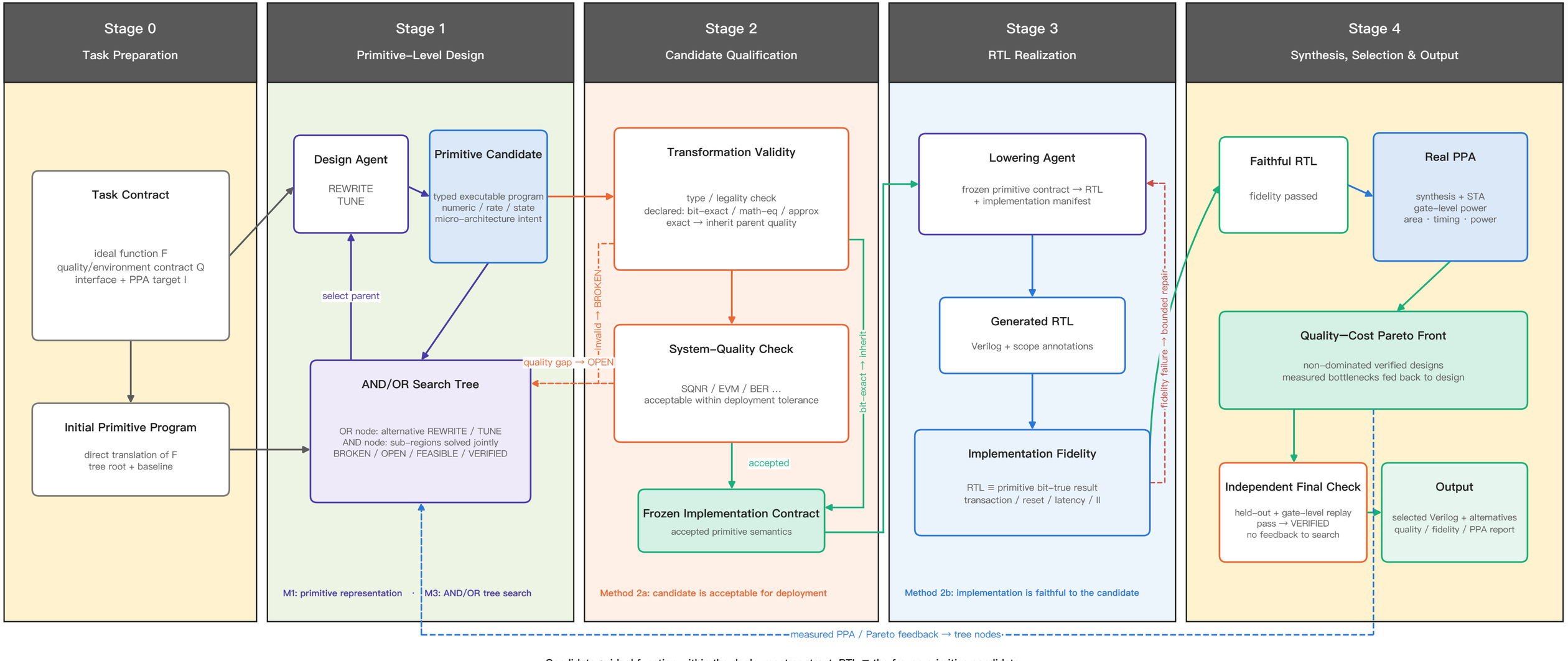
Stage 0
Task Preparation
Stage 1
Primitive–Level Design
Stage 2
Candidate Qualification
Stage 3
RTL Realization
Stage 4
Synthesis, Selection & Output
Task Contract
ideal function F
quality/environment contract Q
interface + PPA target I
Initial Primitive Program
direct translation of F
tree root + baseline
Design Agent
REWRITE
TUNE
Primitive Candidate
typed executable program
numeric / rate / state
micro–architecture intent
select parent
AND/OR Search Tree
OR node: alternative REWRITE / TUNE
AND node: sub–regions solved jointly
BROKEN / OPEN / FEASIBLE / VERIFIED
M1: primitive representation · M3: AND/OR tree search
Transformation Validity
type / legality check
declared: bit–exact / math–eq / approx
exact → inherit parent quality
invalid → BROKEN
quality gap → OPEN
System–Quality Check
SQNR / EVM / BER …
acceptable within deployment tolerance
accepted
Frozen Implementation Contract
accepted primitive semantics
bit–exact → inherit
Method 2a: candidate is acceptable for deployment
Lowering Agent
frozen primitive contract → RTL
+ implementation manifest
Generated RTL
Verilog + scope annotations
Implementation Fidelity
RTL ≡ primitive bit–true result
transaction / reset / latency / II
fidelity failure → bounded repair
Method 2b: implementation is faithful to the candidate
Faithful RTL
fidelity passed
Real PPA
synthesis + STA
gate–level power
area · timing · power
Quality–Cost Pareto Front
non–dominated verified designs
measured bottlenecks fed back to design
Independent Final Check
held–out + gate–level replay
pass → VERIFIED
no feedback to search
Output
selected Verilog + alternatives
quality / fidelity / PPA report
measured PPA / Pareto feedback → tree nodes
Candidate ≈ ideal function within the deployment contract; RTL ≡ the frozen primitive candidate.